

Programme	B. Sc. Mathematics Honours			
Course Code	MAT3FV109(1)			
Course Title	HISTORY OF MATHEMATICS			
Type of Course	VAC			
Semester	III			
Academic Level	200 - 299			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	3	3	-	45
Pre-requisites	Aptitude for Mathematics and its History.			
Course Summary	The course goes into the philosophy of mathematics, modern axiom methods, controversies in set theory around axiom of choice, its implications and various philosophical alternative approaches to the foundations of mathematics.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Analyse Key Mathematical Theorems and Concepts from Ancient to Early Modern Times	An	C	Internal Exam/ Assignment/ Seminar/ Viva / End Sem Exam
CO2	Evaluate and Compare Methods of Addressing Infinity and Large Cardinal Numbers	E	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Ensure students gain a comprehensive understanding of the historical development and foundational concepts of mathematics	An	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	Mathematics & Its History, 3 rd Edition, John Stillwell, Springer (2010) ISBN: 978-1-4419-6052-8.			
Module	Unit	Content	Hrs (36+9)	Ext. Marks (50)
I	Ancient Origins & Foundations		9	Min 10
	Quick Review of Ancient Mathematics			
	1	Chapter 1: Pythagoras Theorem		
	2	Chapter 2: Greek Geometry		
	3	Chapter 3: Greek Number Theory		
	Infinity in Greek Mathematics – Chapter 4			
	4	Section 4.1, 4.2-Fear of Infinity, Eudoxus’ Theory of Proportions		
	5	Section – 4.3, 4.4-The Method of Exhaustion, Area of a Parabolic Segment		
	Sets & Logic – Chapter 24			
	6	Sections 24.1, 24.2, 24.4- Sets, Ordinals, Axiom of Choice & Large Cardinals		
	7	Section 24.3- Measure		
	8	Section 24.5-The Diagonal Argument		
	Biographical Notes: Pythagoras, Euclid, Diophantus, Archimedes			
II	Calculus – Chapter 9		9	Min 10
	9	Section 9.1, 9.2-What is Calculus, Early Results on Areas & Volumes		
	10	Section 9.3-Maxima, Minima & Tangents		
	11	Section 9.4-The <i>Arithmetica Infinitorum</i> of Wallis		
	12	Section 9.5-Newton’s Calculus of Series		
	13	Section 9.6-The Calculus of Leibnitz		

	Biographical Notes: Wallis, Newton & Leibnitz			
III	Algebraic Equations & Numbers		9	Min 10
	Polynomial Equations – Chapter 6			
	14	Section 6.1, 6.2- Algebra, Linear Equations & Elimination		
	15	Section 6.3, 6.4 Quadratic Equations, Quadratic Irrationals		
	16	Section 6.5-The Solution of the Cubic		
	17	Section 6.6-Angle Division		
	18	Section 6.7-Higher Degree Equations		
	Biographical Notes: Tartaglia, Cardano & Viete			
	Complex Numbers – Chapter 14			
	19	Section 14.1, 14.2, 14.3- Impossible Numbers, Quadratic & Cubic Equations		
	20	Section 14.4- Wallis’ Attempt at Geometric Representation		
	21	Section 14.5, 14.6- The Fundamental Theorem of Algebra, The Proofs of d’Alembert & Gauss		
	Biographical Notes: d’Alembert			
IV	Topology – Chapter 22		10	Min 10
	22	Section 22.1, 22.2- Geometry & Topology, Polyhedron Formulas of Descartes & Euler		
	23	Section 22.3-The Classification of Surfaces		
	24	Section 22.4- Descartes & Gauss-Bonnet		
	25	Section Euler 22.5-Characteristic & Curvature		
	26	Section 22.7, 22.8- The Fundamental Group, The Poincare Conjecture		
	Biographical Notes: Poincare			
V	Open Ended Module		9	
	1	Hypercomplex Numbers – Chapter 20		

	2	Number Theory in Asia – Chapter 5		
	3	Mechanics – Chapter 13		
	4	Complex Numbers & Functions – Chapter 16		
	5	Non-Euclidean Geometry – Chapter 18		
	6	Group Theory – Chapter 19		

References:

1. Mathematics, The Queen & Handmaiden of Sciences, E. T. Bell, McGraw Hill.
2. Men of Mathematics, E. T. Bell, Simon & Schuster, 1986.
3. What is Mathematics?, Richard Courant & Herbert Robbins,
4. History of Mathematics, 7th Edition, David M. Burton, McGraw Hill.
5. Mathematics In India, Kim Plofker, Princeton University Press, 2009.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	1	0	3	2	2	0	3	2	1
CO 2	3	2	1	0	2	1	2	0	2	1	0
CO 3	1	1	0	0	3	2	2	0	3	2	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓